

Predictive Maintenance For Auto Ancillary Manufacturing



\$7.3 Billion

Market Cap As Of 2025

20 Mfg Units

74 Manufacturing Plants
Worldwide

6,000+ Employees

Across All Of The Business
Verticals

Introduction

A leading Indian multinational corporation, founded in 1992, renowned for its expertise in automotive solutions, manufacturing components for the automotive and transportation industries.

Generating significant annual revenue, the company focuses on innovation and cutting-edge manufacturing technologies to stay competitive and maintain a strong presence in the global market.

At several key facilities, reliance on manual maintenance for critical rotating assets led to unplanned downtime, prompting plant teams to seek advanced solutions.

This study explores how the organization transformed its maintenance practices by implementing IoT-enabled condition monitoring and predictive maintenance, significantly improving operational efficiency and reducing downtime.

Background

At a large manufacturing facility, equipment performance data was collected using handheld sensors and analyzed by experts to identify defects.

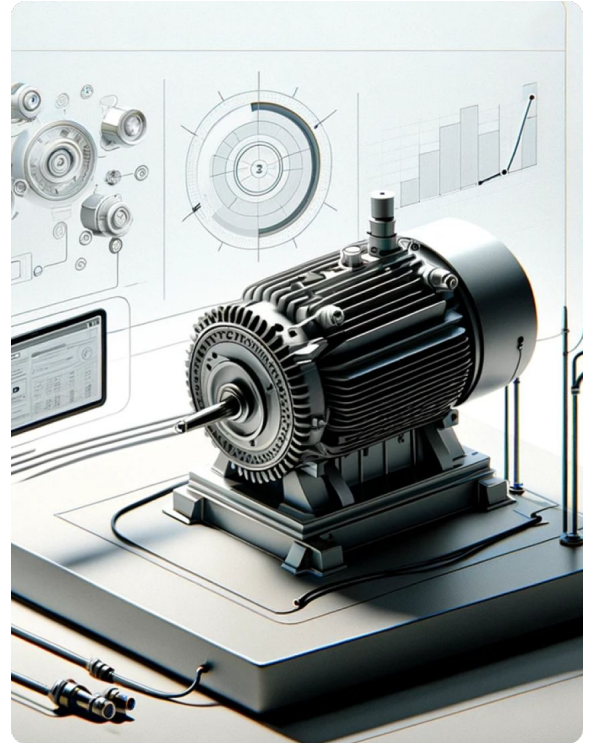
However, this method was inadequate due to the high number of critical rotating assets, infrequent data collection, and reliance on manual analysis, limiting effective detection and prevention of equipment failures.



Current Challenges

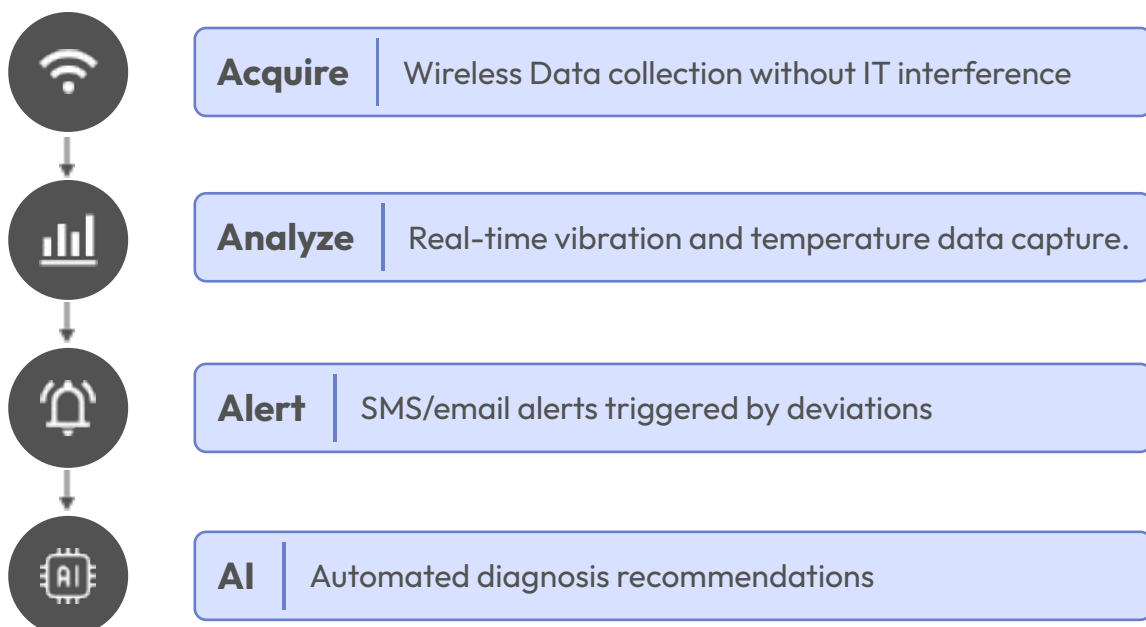
Manual vibration analysis of hundreds of rotating machines is a highly time consuming task, that has various challenges as follows:

1. **Time consuming analysis:** Manually capturing data using handheld instrumentation across hundreds of machines, followed by the assessment of readings, was time-consuming and delayed repairs.
2. **Safety Hazard:** Manual data collection across a steel manufacturing plant poses substantial safety risks and concerns for auditors, even with all the protective equipment in place.
3. **Productivity Losses:** Due to delays in diagnosis, teams faced unplanned downtimes caused by the failure of critical rotating machinery, resulting in significant productivity losses.
4. **Need for real-time visibility:** It was evident that there was a need for a real-time solution that could automatically collect data and offer remote diagnostics to ensure timely repairs.



Solution : IoT And AI Enabled Condition Monitoring And Predictive Maintenance

In response to these challenges, Faclon identified the following critical assets and implemented a highly strategic deployment on these assets



Key Machines

Machine Type	Function
Hot Water Rinse	Rinse Components With Hot Water.
Pre Degreasing	Remove Heavy Grease And Dirt.
Degreasing	Eliminate Oils And Residues.
Ro Water Rinse	Pure Water Rinsing To Remove Contaminants.
Cleaner Coating	Apply Surface Conditioning Coatings.
Dm Water Pump	Supply Ultra-pure Water.
Oxolin	Apply Anti-rust Coatings.
Hwg Pump	Deliver Hot Water For Processes.
Sludge Pit Pump	Remove Sludge And Debris.
Pt Conveyor Motor	Drive Conveyor Through Stages.
Lp Hac Blower	Provide Airflow For Drying/chemical Baths.
Ahu Blower	Control Air Temperature And Humidity.
Wdo Hac Blower	Support Waste/wet Dip Operations.
Pbo Hac Blower	Air Circulation For Drying/blow-off.

Hardware Methodology

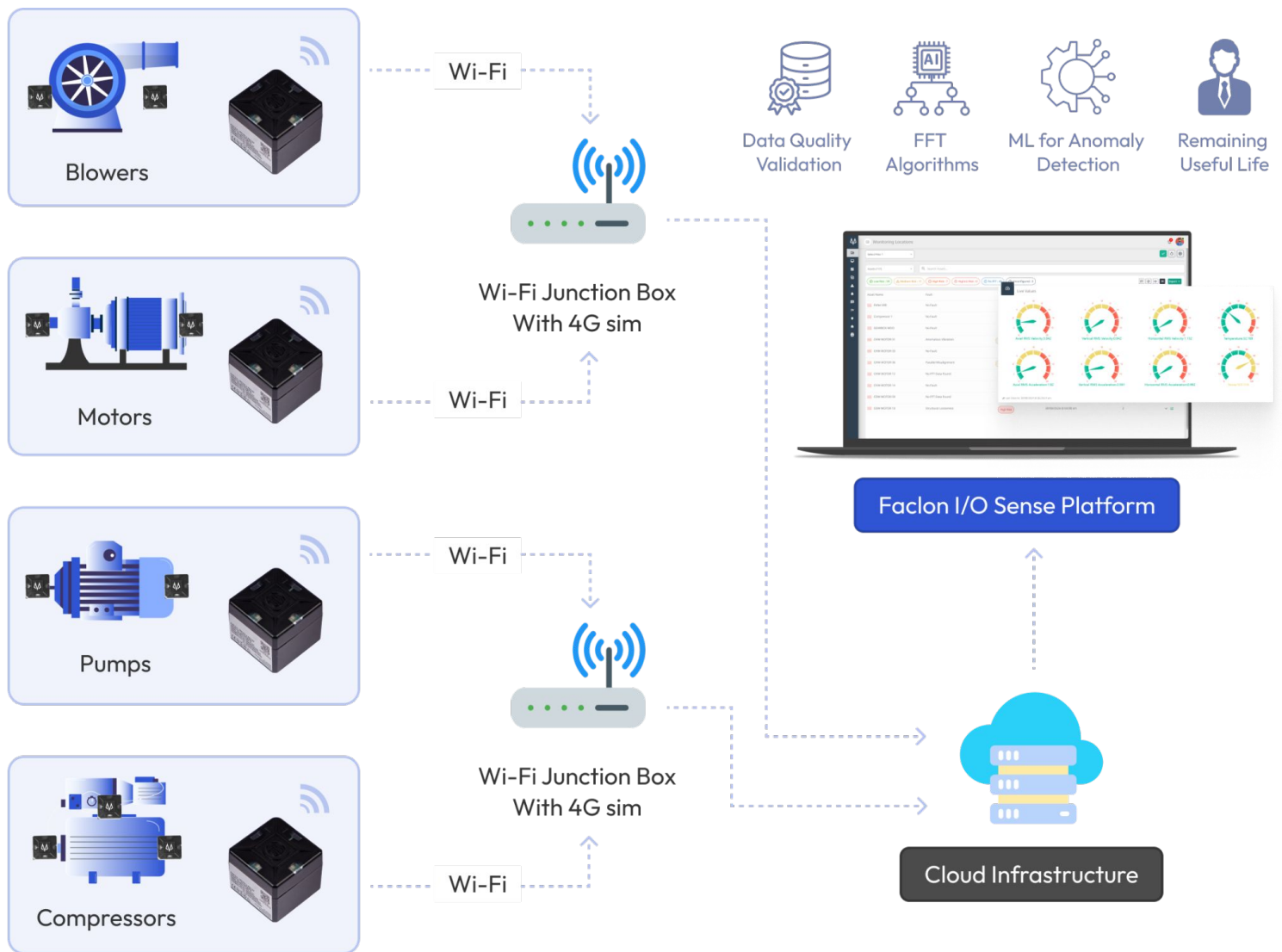
The first step involved sensorization of critical rotating assets such as motors, fans, pumps and gearboxes using our MEMS based Triaxial Vibration Sensors.

40+

Sensors Deployed

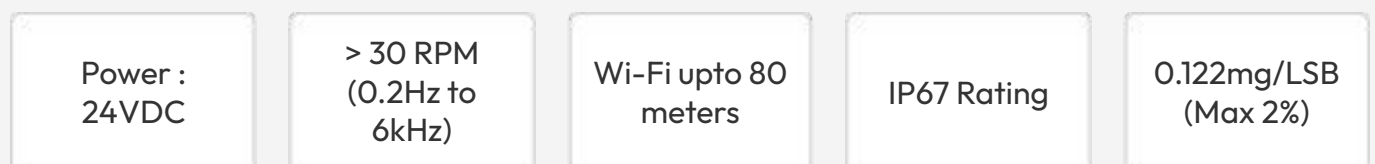
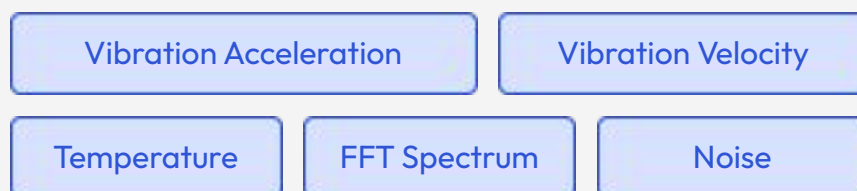
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Machine Digitized



Sensor Specification

MEMS based Triaxial Sensor transmit data wirelessly via Wi-Fi to the Junction boxes. These junction boxes directly transmit the data to the I/O Sense Cloud platform via 4G. Each sensor collects the following parameters:



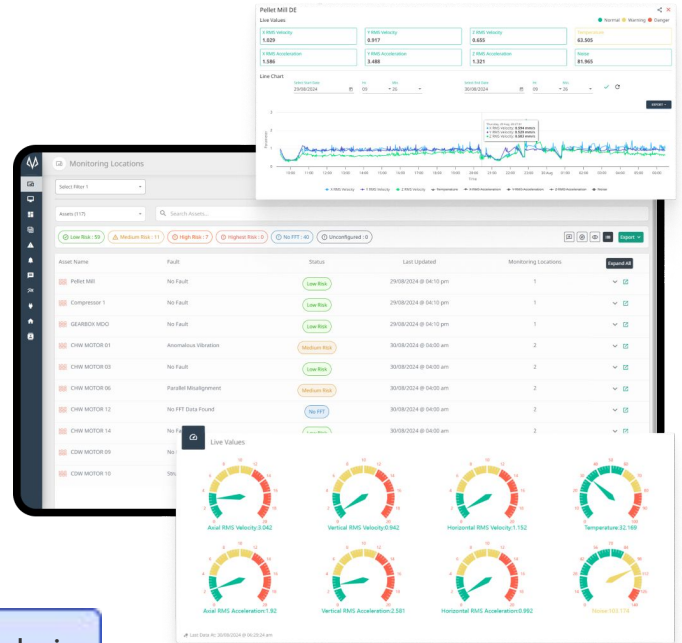
Platform Methodology

Analyze

Once all sensor data is onboarded on the I/O Sense Platform, based different asset types and configurations, the fault frequencies are automatically adjusted and ISO benchmarks for Vibration Velocity and Acceleration.

The master dashboard, based on this benchmark, displays the health status of every connected asset.

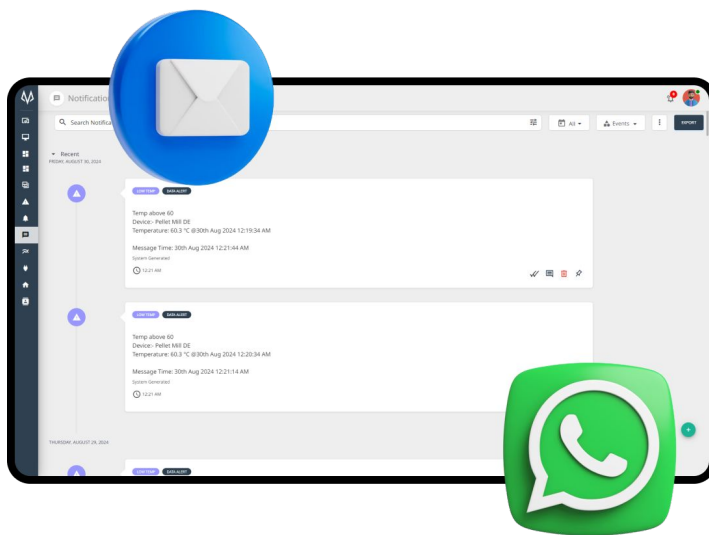
Upon further inspection, users can also review plotted graphs to verify deviations in trends.



FFT Analysis

Noise Analysis

Trend Analysis

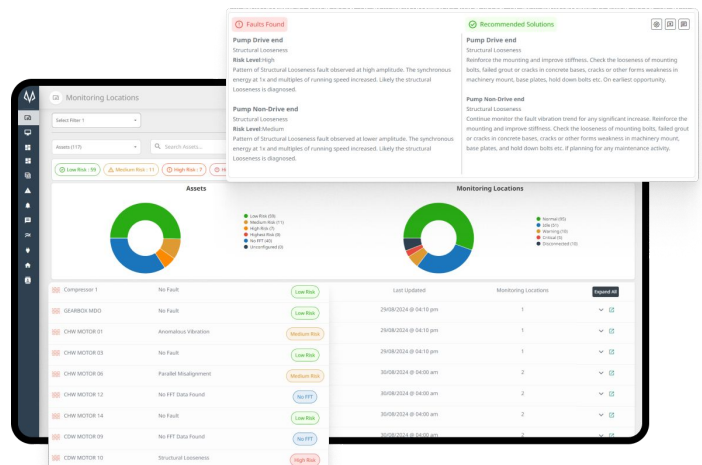


Alert

As soon as a deviation in the machine's vibration velocity and acceleration from ISO Standards is observed, alerts are sent to relevant personnel via SMS, Email or WhatsApp.

Artificial Intelligence

The platform's machine learning algorithms further analyze FFT spectrums generated every 30 minutes and give a diagnostic inference every 24 hours.



Implementation Success Story

75%

Unplanned
downtime reduced

12

Critical defects
detected till date

480

Man-days saved

18%

Reduced spends on
maintenance



Asset Name : Blower Motor

Total Number of Sensors Installed : 2

Sensor location : Motor Drive End, Motor Non-Drive end.

Fault Diagnosed : Alignment Issues has been Observed via Sensor - Motor DE & NDE.

Final Action : Client aligned the shaft as required.

Savings : Reduced Maintenance cost of further damages in Gears which is very costly in Gearbox if not intervened in time

Before



After



As we can see, the overall body vibrations have reduced significantly after corrective maintenance, resulting in healthy machine operations

Conclusion

An industry 4.0 roadmap when implemented with well-thought outcomes in mind can yield transformational outcomes:



Real-time Alerts

Shifting from manual to real-time asset health monitoring has notably reduced correction times, effectively mitigating million-dollar losses from unexpected downtime.



Comprehensive Analysis

The feature-rich system enables comprehensive analysis, improving decision-making and predictions for asset maintenance and performance.



Enhanced Productivity

The system significantly reduces manpower needs and minimizes the risk for personnel collecting data in hazardous locations.



ROI Realization

ROI is typically realized within 1 to 1.5 years, making the system a cost-effective investment with quick financial returns.



This implementation study demonstrates benefits of IoT enabled predictive maintenance and its impact in the auto-manufacturing industry



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